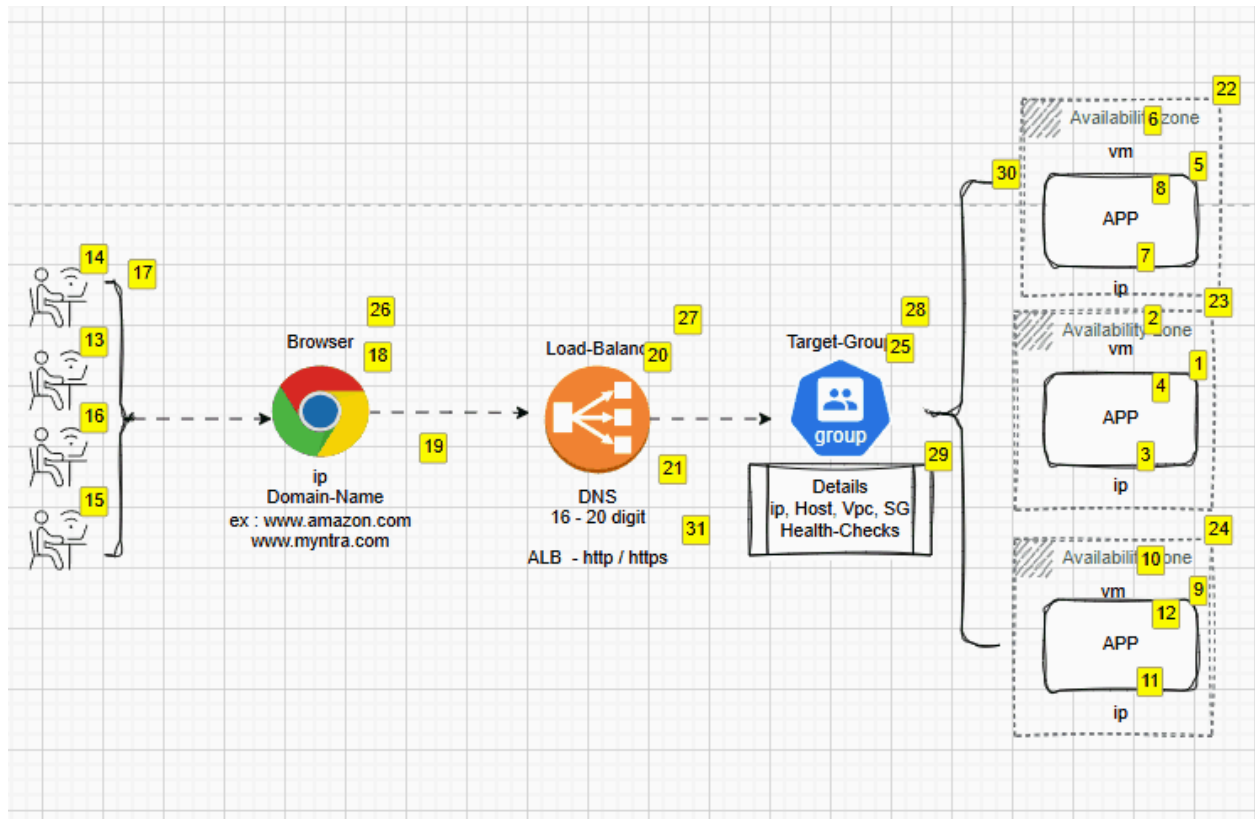




🌐 What is an Application Load Balancer (ALB)?

An **Application Load Balancer (ALB)** is a **Layer 7** (Application Layer) load balancer provided by AWS Elastic Load Balancing. It is designed to handle **HTTP and HTTPS** traffic and offers **advanced request routing**, including:

- **Content-based routing** (path, host, query string, headers)
- **WebSocket and HTTP/2** support
- **Container-based applications** (like with ECS or EKS)
- **Authentication and SSL termination**

Components of an Application Load Balancer

Here are all the **core components** of an ALB:

1. Listeners

- A **listener** is a process that checks for connection requests.
- Each listener is configured with:
 - **Protocol:** HTTP or HTTPS
 - **Port:** Typically 80 (HTTP) or 443 (HTTPS)
- A listener **forwards requests to rules** for routing to target groups.

Example:

Listener on port 80 -> Forwards to rules

Listener on port 443 -> SSL termination + forwards to rules

2. Listener Rules

- Listener rules determine **how incoming requests are routed**.
- Each rule consists of:
 - **Conditions:** e.g., Host header, Path pattern (*/api/**), HTTP headers, query string
 - **Actions:** Forward to a target group, redirect, fixed response, or authenticate

Rule Example:

IF Host is api.example.com AND path is /v1/*
→ Forward to Target Group A

3. Target Groups

- A **target group** is a logical group of targets (EC2 instances, IPs, Lambda functions, containers).
- You **attach health checks** and routing rules to a target group.
- A single ALB can route traffic to **multiple target groups** based on listener rules.

Target Types:

- **instance**: Routes to EC2 instances
 - **ip**: Routes to private IPs (good for on-premise or non-AWS targets)
 - **lambda**: Routes to AWS Lambda functions
 - **alb**: Routes to another ALB (chaining)
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4. Targets

- These are the **actual destinations** for incoming traffic.
- Can be:
 - EC2 instances
 - ECS containers

- IP addresses
- Lambda functions

Targets must be registered with a **target group**.

5. Availability Zones (AZs)

- ALBs are **multi-AZ**, providing **high availability and fault tolerance**.
 - You can choose one or more AZs when creating the ALB.
 - AWS automatically distributes traffic across healthy targets in these zones (especially with **cross-zone load balancing** enabled).
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6. Health Checks

- ALB uses health checks to determine if a target is **healthy and available**.
- If a target fails health checks repeatedly, it is **removed from routing** until it recovers.



Configurable:

- Protocol (HTTP/HTTPS)
 - Path (/health)
 - Interval, timeout, and threshold
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7. Security Groups

- ALB uses **security groups** to control inbound/outbound traffic.
 - Typical settings:
 - Allow inbound traffic on port 80/443
 - Restrict outbound access to known target IPs or ranges
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8. SSL Termination (HTTPS Listener)

- ALB can terminate SSL (HTTPS) traffic.
 - You must:
 - Use an **SSL certificate** (from ACM or uploaded manually)
 - Configure **HTTPS listener**
 - Offloads encryption from backend servers (performance benefit)
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9. WAF (Web Application Firewall) Integration

- ALB integrates with **AWS WAF** to protect web apps from:
 - SQL injection
 - Cross-site scripting (XSS)
 - Bot traffic
 - Rate limiting

10. Access Logs

- ALB can log **detailed request information** to an S3 bucket:
 - Client IP
 - Target and response time
 - Status codes
 - User-agent
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11. Host-based and Path-based Routing

- ALB excels at advanced routing:
 - **Host-based:** `api.example.com`, `shop.example.com`
 - **Path-based:** `/api/`, `/admin/`

This enables **microservice** and **container-based architectures**.

✅ Summary of ALB Capabilities

Feature	ALB Support
Layer 7 routing (HTTP/HTTPS)	✅
Host- and path-based routing	✅
WebSocket & HTTP/2	✅
Container support (ECS, EKS)	✅

SSL termination	✓
AWS Lambda target support	✓
IP address target support	✓
Authentication via Cognito/OIDC	✓
WAF integration	✓

Typical Use Case Architecture

Client → ALB (HTTPS Listener) → Listener Rule → Target Group → ECS Container or EC2
